Dynamic Stability Index for Early Learning Rate Selection in Deep Neural Networks

Abstract

Learning rate selection is a central problem in deep neural network training because it directly affects convergence, numerical stability, and final model performance. Although exhaustive validation can identify competitive configurations, its exploration cost may become substantial when multiple alternatives must be evaluated. This work proposes an early selection criterion based on the initial training dynamics. The formulation relies on a Dynamic Stability Index (DSI), which synthesizes empirical volatility, oscillation, normalized descent, and numerical instability into a single operational measure.

The empirical evaluation is organized at several levels and includes a controlled main experiment, comparisons against reference methods, validation across multiple datasets, and real-world applications. The results show that the proposed criterion can identify competitive learning rates with substantially lower search cost than grid search in diverse scenarios. In particular, the method achieves substantial reductions in time and in the number of exploratory batches while maintaining, in several cases, final performance close to exhaustive validation. In addition, sensitivity and robustness analyses suggest that the signal captured by the index remains stable under different configurations.

Overall, the evidence indicates that early training dynamics can be effectively exploited to build simple, interpretable, and computationally efficient learning-rate selection rules.

Keywords: learning rate selection, hyperparameter optimization, early training dynamics, dynamic stability, deep neural networks

1. Introduction

Learning rate selection is one of the most sensitive problems in deep neural network training because of its direct influence on convergence speed,

numerical stability, and the final quality of the obtained solution. In practice, values that are too small may slow down learning, whereas excessively large values may induce oscillations, instability, or divergence. For this reason, learning-rate choice occupies a central role in hyperparameter optimization, especially in scenarios where training multiple configurations is costly.

Although general strategies exist to reduce exploration cost, there is still a clear need for specific, simple, and interpretable mechanisms for early learning-rate selection. In particular, accumulated evidence suggests that the early stages of training contain useful signals about the quality of the optimization regime, but this evidence is not always translated into operational rules that can be directly applied to a discrete grid of candidate values.

In this context, the present work proposes an early selection criterion based on the initial loss dynamics. The proposal is organized around a Dynamic Stability Index (DSI), which summarizes empirical volatility, oscillation, normalized descent, and numerical instability into a single measure, together with a selection rule that prioritizes relatively stable configurations with effective early progress.

The objective of this article is to evaluate whether a signal built from early training dynamics can be used to select competitive learning rates with substantially lower exploration cost than exhaustive validation. In this sense, the central hypothesis is that the early stages of training contain sufficient information to identify favorable configurations without fully training all alternatives.

To examine this hypothesis, we develop a multi-level empirical evaluation that includes a controlled main experiment, comparisons against reference methods, validation across multiple datasets, multi-seed robustness analysis, and applications in heterogeneous real-world scenarios. Within this framework, the main contribution of the paper is threefold: to propose a compact measure for characterizing early training trajectories, to formulate an operational selection rule based on that measure, and to document its behavior under a reproducible experimental setup supported by a public repository and supplementary material.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the proposed methodology. Section 4 describes the experimental design. Section 5 reports the results and discussion. Finally, Section 6 summarizes the main conclusions and outlines future work.

2. Related Work

Learning-rate selection has been addressed from different perspectives within hyperparameter optimization, adaptive budget allocation, and the analysis of training dynamics.

Within the general hyperparameter optimization framework, Bergstra and Bengio showed that random search can be more efficient than exhaustive grid search when only a small subset of hyperparameters accounts for most of the performance variation. Later, the increasing computational cost of model selection fostered the development of multi-fidelity approaches aimed at comparing configurations under partial budgets.

In this line, Hyperband combined adaptive resource allocation with early elimination of unpromising configurations, whereas BOHB integrated this logic with Bayesian optimization to obtain a more robust and scalable procedure in expensive search spaces. More recently, review studies have confirmed that multi-fidelity optimization has become a central strategy for reducing the cost of model tuning in contemporary machine learning.

In parallel, several works started to exploit the partial information generated during training itself. Under this perspective, learning-curve extrapolation showed that exploration can be accelerated without training every configuration to completion. However, these approaches often rely on auxiliary models or on budget-allocation schemes that are more complex than a direct selection rule.

In the specific case of learning rates, Smith showed that the early evolution of the loss can be used to delimit plausible search ranges and guide scheduling policies. More recently, training dynamics have also been studied as an object of analysis in their own right. In particular, work on the so-called edge of stability showed that some trajectories may exhibit persistent oscillations without necessarily implying pathological behavior, but rather a structured dynamical regime associated with the interaction between step size and local geometry. In the same direction, Kalra and Barkeshli showed that early dynamics depend systematically on the learning rate, network depth, and width, reinforcing the idea that early training stages contain relevant information for characterizing optimization quality.

Taken together, prior literature shows substantial progress in hyperparameter optimization, multi-fidelity strategies, exploitation of partial training information, and dynamical analysis of learning. However, there is still a clear space for learning-rate selection criteria that are simultaneously early,

interpretable, and computationally lightweight, and that can be constructed directly from observable signals in a short training window without relying on complex auxiliary models or extensive exploration procedures.

3. Proposed Methodology

3.1. Problem Formulation

Let

$$\mathcal{L} = \{\eta_1, \eta_2, \ldots, \eta_m\} \tag{1}$$

be a discrete set of candidate learning rates. The goal is to select, early and efficiently, a configuration $\eta^* \in \mathcal{L}$ that preserves competitive final performance without requiring full training of all alternatives. If $J(\eta)$ denotes the final performance associated with learning rate η and $C(\eta)$ its exploration cost, the problem may be understood as the search for a configuration that offers a favorable trade-off between final quality and selection cost.

Unlike approaches based exclusively on exhaustive validation or on general hyperparameter-optimization strategies, the proposed method adopts a perspective centered on early training dynamics. Instead of estimating the quality of each configuration from full convergence, the method exploits observable signals in a short training window to distinguish relatively stable trajectories, oscillatory trajectories, and numerically unstable trajectories.

3.2. Early Exploration of Training Trajectories

For each $\eta \in \mathcal{L}$, a short exploratory phase is executed and the loss sequence

$$\mathcal{L}(\eta) = \{l_1(\eta), l_2(\eta), \ldots, l_T(\eta)\} \tag{2}$$

is recorded, where T depends on the budget effectively consumed during exploration. During the exploratory phase intended for index computation, in addition to the loss, the global gradient norm, the global weight norm, the number of processed batches, the elapsed time, and the possible occurrence of numerical instabilities are also recorded.

These complementary records are retained for traceability and diagnostic purposes, whereas the index itself is computed directly from the loss trajectory and the numerical-instability indicator. A trajectory is considered unstable if non-finite losses, non-finite gradients, or non-finite parameters appear during exploration. In such cases, the corresponding configuration

is marked as divergent, and this information is explicitly retained for the selection stage.

In the exploratory phase used for DSI computation, the budget is not fixed solely in terms of nominal epochs. To guarantee a minimum number of useful observations, the number of dynamic epochs is automatically adjusted as

$$E_{\text{dyn}} = \max\left(1, \left\lceil \frac{M}{\min\left(|\mathcal{B}_{\text{train}}|, B\right)} \right\rceil\right), \tag{3}$$

where M denotes the minimum number of required valid observations, $|\mathcal{B}_{\text{train}}|$ is the number of batches per epoch, and B is the maximum number of batches allowed per configuration. In the main configuration, $M = 8$ and $B = 30$.

This stage transforms learning-rate selection into a problem of comparing early trajectories. The underlying hypothesis, consistent with prior work on early training dynamics, is that the first iterations contain sufficient information to characterize qualitatively different optimization regimes.

3.3. Proposed Criterion: Dynamic Stability Index

The core of the proposal is the Dynamic Stability Index (DSI), defined as

$$\text{DSI}(\eta) = \alpha V(\eta) + \beta O(\eta) - \gamma D(\eta) + \delta P(\eta), \tag{4}$$

where $V(\eta)$ denotes empirical loss volatility, $O(\eta)$ empirical oscillation, $D(\eta)$ normalized descent, and $P(\eta)$ a penalty for numerical instability. This formulation translates the previous intuition into an operational early-selection criterion by combining stability, oscillation, early progress, and numerical instability into a single measure.

In the main configuration, we use $\alpha = 1.0$, $\beta = 0.75$, $\gamma = 1.25$, and $\delta = 2.0$. This parametrization is not claimed to be universally optimal, but rather to provide an operational setting that slightly prioritizes early loss progress, maintains control over volatility and oscillation, and explicitly penalizes numerically unstable trajectories. Its influence is later evaluated through ablation and sensitivity studies over a predefined grid of weights.

The index components are computed on the filtered series of finite losses,

$$\widetilde{\mathcal{L}}(\eta) = \{\tilde{l}_1(\eta), \tilde{l}_2(\eta), \ldots, \tilde{l}_{\tilde{T}}(\eta)\}, \tag{5}$$

obtained after removing non-finite observations. The index is considered undefined when the filtered series does not allow valid computation of its main components. In particular, volatility and oscillation require a minimum number of finite observations, whereas normalized descent requires at least two valid points.

3.3.1. Empirical Volatility

First, relative differences between consecutive losses are defined as

$$r_t(\eta) = \frac{\left|\tilde{l}_t(\eta) - \tilde{l}_{t-1}(\eta)\right|}{\left|\tilde{l}_{t-1}(\eta)\right| + \varepsilon}, \qquad t = 2, \ldots, \tilde{T}, \tag{6}$$

where $\varepsilon > 0$ is a small constant to avoid division by zero. Volatility is then defined as

$$V(\eta) = \frac{1}{\tilde{T} - 1} \sum_{t=2}^{\tilde{T}} \log\bigl(1 + r_t(\eta)\bigr). \tag{7}$$

This component captures abrupt relative variations in the loss trajectory and penalizes configurations with irregular behavior during the exploratory phase.

3.3.2. Empirical Oscillation

Let

$$\Delta_t(\eta) = \tilde{l}_t(\eta) - \tilde{l}_{t-1}(\eta), \qquad t = 2, \ldots, \tilde{T}. \tag{8}$$

Empirical oscillation is computed as the proportion of sign changes between consecutive differences:

$$O(\eta) = \frac{1}{\tilde{T} - 2} \sum_{t=3}^{\tilde{T}} \mathbb{1}\left[\Delta_t(\eta)\Delta_{t-1}(\eta) < 0\right]. \tag{9}$$

This term quantifies how often the loss alternates between local increases and decreases, providing information about the oscillatory character of the trajectory.

3.3.3. Normalized Descent

Normalized descent measures the net progress achieved between the beginning and the end of the exploratory phase:

$$D(\eta) = \frac{\tilde{l}_1(\eta) - \tilde{l}_{\tilde{T}}(\eta)}{\left|\tilde{l}_1(\eta)\right| + \varepsilon}. \tag{10}$$

This term favors configurations that are not only relatively stable, but also produce an effective reduction of the loss at early stages.

3.3.4. Instability Penalty

The instability penalty is defined as

$$P(\eta) = \begin{cases} 1, & \text{if the trajectory exhibits divergence or non-finite values,} \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

Whenever $P(\eta) = 1$, we directly assign

$$\mathrm{DSI}(\eta) = +\infty, \tag{12}$$

so that the configuration is excluded from the effective comparison. Therefore, although the general formulation includes the term $\delta P(\eta)$, in the implementation used here the penalty operates as a discard criterion.

Overall, the DSI privileges trajectories with low volatility, low oscillation, and effective loss descent, while explicitly penalizing numerically unstable behaviors.

3.4. Selection Rule

Final selection is not based on the absolute minimum of the index, but on a more robust rule that combines relative stability and descent capacity. Let

$$\mathcal{L}_{\text{fin}} = \{\eta \in \mathcal{L} : \mathrm{DSI}(\eta) \text{ is finite and } D(\eta) \text{ is defined}\} \tag{13}$$

be the set of valid configurations. Over this subset, the q-quantile of the index values is computed:

$$Q_q = \mathrm{Quantile}_q\left(\{\mathrm{DSI}(\eta) : \eta \in \mathcal{L}_{\text{fin}}\}\right). \tag{14}$$

The stable region is then defined as

$$\mathcal{R}_q = \{\eta \in \mathcal{L}_{\text{fin}} : \mathrm{DSI}(\eta) \leq Q_q\}. \tag{15}$$

The final decision rule is

$$\eta^* = \begin{cases} \arg\min\limits_{\eta \in \mathcal{L}_{\text{fin}}} \mathrm{DSI}(\eta), & \text{if } \mathcal{R}_q = \varnothing, \\ \arg\max\limits_{\eta \in \mathcal{R}_q} D(\eta), & \text{otherwise.} \end{cases} \tag{16}$$

The main version of the method uses $q = 0.33$. This rule reduces the risk that the decision is dominated by a single extreme index value and first restricts the comparison to a region of relatively stable configurations, then selecting the one with the greatest early progress within that region.

3.5. Algorithm

Algorithm 1. DSI-based dynamic learning-rate selection.

Input: candidate set $\mathcal{L} = \{\eta_1, \eta_2, \ldots, \eta_m\}$, minimum number of valid observations M, maximum exploration budget B, weights $(\alpha, \beta, \gamma, \delta)$, and quantile q.

Output: selected learning rate η^*.

1. Determine the number of exploratory epochs required to guarantee at least M valid observations while respecting the maximum of B batches per configuration.
2. For each $\eta \in \mathcal{L}$, execute a short exploratory trajectory.
3. Record the loss series $\mathcal{L}(\eta)$, the global gradient and weight norms, the elapsed time, and the number of processed batches.
4. Detect whether the trajectory exhibits non-finite losses, non-finite gradients, or non-finite parameters.
5. Compute $V(\eta)$, $O(\eta)$, $D(\eta)$, and $P(\eta)$.
6. Compute $\mathrm{DSI}(\eta) = \alpha V(\eta) + \beta O(\eta) - \gamma D(\eta) + \delta P(\eta)$.
7. If $P(\eta) = 1$, assign $\mathrm{DSI}(\eta) = +\infty$.
8. Construct the set $\mathcal{L}_{\mathrm{fin}}$ of valid configurations with finite DSI and defined descent.
9. Compute the quantile Q_q over the valid index values.
10. Define the stable region $\mathcal{R}_q = \{\eta \in \mathcal{L}_{\mathrm{fin}} : \mathrm{DSI}(\eta) \leq Q_q\}$.
11. If $\mathcal{R}_q$ is empty, select the configuration with minimum finite DSI.
12. Otherwise, select the configuration with maximum normalized descent within $\mathcal{R}_q$.
13. Return η^*.

3.6. Ablation and Sensitivity

To evaluate the relative contribution of each index component, we perform an ablation study in which, in addition to the full DSI, we compute isolated volatility, isolated oscillation, the descent component taken alone as a reward term, the volatility–oscillation combination, and the instability penalty. This analysis makes it possible to examine whether the patterns observed in the composite criterion are consistent with simpler partial signals.

Complementarily, a sensitivity analysis is conducted over $(\alpha, \beta, \gamma, \delta)$ by recalculating the index on a predefined grid of combinations. In the implementation used here, we explore $\alpha \in \{0.5, 1.0, 1.5\}$, $\beta \in \{0.5, 0.75, 1.0\}$, $\gamma \in \{1.0, 1.25, 1.5\}$, and $\delta \in \{1.0, 2.0, 3.0\}$. In addition, the sensitivity of the selection rule is evaluated by comparing the main criterion with variants based on different quantiles and alternative rules. These tests make it possible to analyze the robustness of the method beyond a single parametrization.

4. Experimental Design

4.1. Datasets and Splits

The empirical evaluation is organized into three levels: controlled vision experiments, multi-dataset validation, and real-world applications.

The vision experiments use MNIST, FashionMNIST, CIFAR-10, and CIFAR-100. For the search phase, a random subset of the official training set is taken and divided into search-training and validation subsets using an 80/20 split. The official test set is reserved for final evaluation. In the main configuration, we use 10,000 examples for MNIST, 8,000 for FashionMNIST, 10,000 for CIFAR-10, and 10,000 for CIFAR-100. In the multi-dataset study, MNIST and FashionMNIST are additionally evaluated with subsets of 6,000 observations.

As tabular references, we use Wine, Breast Cancer, and Digits. In these cases, a stratified 70/30 split is first applied between full training and test data, followed by a second stratified 80/20 split inside the training set to obtain search and validation subsets. Variables are normalized with StandardScaler.

The real-world applications include Raisin and NASA-CMAPSS FD001. For Raisin, a stratified 70/15/15 split is used for training, validation, and test data. For NASA-CMAPSS, the split is performed by engine identifier, also in a 70/15/15 proportion, to avoid information leakage across cycles

belonging to the same engine. In the latter case, the problem is formulated as a binary risk-classification task, where the positive class is defined by RUL ≤ 30. Because of class imbalance, class weights are used during training. StandardScaler is used for Raisin, whereas MinMaxScaler is used for the selected NASA-CMAPSS variables.

4.2. Architectures and Training Configurations

The evaluation includes architectures of different complexity in order to study the behavior of the criterion under heterogeneous regimes. For MNIST, a deep MLP with hidden layers of size 512, 256, 128, and 64 is used. For FashionMNIST, a lightweight CNN with two convolutional blocks followed by a fully connected layer is employed. For CIFAR-10 and CIFAR-100, two model families are considered: a compact CNN and a small-scale ResNet-like architecture. In the multi-dataset vision experiments, a simpler MLP with two hidden layers is also used. For the tabular problems and real-world applications, a lightweight MLP with layers of 32 and 16 units is adopted.

The exploratory phase uses the optimizer defined for each experiment. In the main implementation, SGD, Adam, and RMSprop are combined depending on the scenario: SGD in the main MNIST experiment, Adam in the FashionMNIST method comparison and in the multi-dataset experiments, and SGD, Adam, and RMSprop in the CIFAR architecture analysis. Adam is used in the real-world applications.

The candidate set of learning rates is defined on a logarithmic grid in the interval $[10^{-4}, 1]$, with 12 points in the main experiments and 10 points in the multi-dataset studies and the real-world applications.

The duration of final training depends on the problem. In the small and tabular datasets, 10 epochs are used; in CIFAR, 20 epochs are used for the CNN and 24 for the ResNet-like architecture; in Raisin, 20 epochs are used; and in NASA-CMAPSS, 12 epochs are used. These settings correspond to those employed in the experimental implementation analyzed in this work.

4.3. Reference Methods

The proposed criterion is compared against three main reference strategies. The first is validation-based grid search, which evaluates all candidate learning rates and selects the one that maximizes the corresponding validation metric. The second is an early-loss sweep, which selects the configuration with the lowest mean initial loss computed over a short window of 16 batches.

The third is random search, which samples four learning rates from the same grid and chooses the best one according to validation.

In some complementary experiments, a hybrid criterion based on standardized scores and baselines with StepLR and CosineAnnealingLR are also included, although the latter are interpreted as auxiliary comparisons rather than as strictly equivalent learning-rate selection methods.

The validation selection metric depends on the problem. In the vision experiments and in most tabular benchmarks, validation accuracy is used. In NASA-CMAPSS, because of the imbalanced nature of the problem, balanced accuracy is used as the selection criterion.

4.4. Evaluation Metrics

The analysis considers both final predictive quality and the cost of the search phase. The main metric in the general experiments is test accuracy. For real-world and imbalanced problems, balanced accuracy, precision, recall, and F_1 score are also reported.

To evaluate selection efficiency, we additionally report total search time, total number of exploratory batches, accuracy per search second, and accuracy per 100 exploratory batches. In the problems where balanced accuracy is reported, its time-normalized and batch-normalized counterparts are also provided. This dual evaluation makes it possible to distinguish methods with similar performance but very different exploration cost.

In the experiments where the relationship between dynamic stability and validation performance is analyzed explicitly, Pearson and Spearman correlations between DSI and the corresponding validation metric are computed.

4.5. Robustness and Statistical Analysis

The stability of the proposed criterion is evaluated through a multi-seed scheme using seeds 7, 21, 42, 84, and 123. Over these repetitions, the dynamic method and the references are compared by paired tests: paired t-test, Wilcoxon signed-rank test, and paired Cohen's d effect size. In addition, descriptive results are summarized in terms of wins, ties, and losses under a practical difference threshold. This set of analyses makes it possible to assess not only mean performance but also consistency under random training variation.

4.6. Implementation, Hardware, and Reproducibility

The experimental implementation was developed in Python using PyTorch, Torchvision, Scikit-learn, and SciPy. The experiments were run in Google Colab and used CUDA acceleration whenever available. To improve reproducibility, a global seed was fixed, data loaders were rebuilt with seed control at each run, and cuDNN was configured in deterministic mode while disabling low-level non-deterministic optimizations.

In addition, loss trajectories, global norms, divergence indicators, search time, number of processed batches, and the main tables and results of the analysis were explicitly recorded. The source code, experimental scripts, used configurations, and derived results will be made available in a public GitHub repository, archived through Zenodo to ensure versioning, persistence, and citability. The digital repository associated with the article also includes the main tables and figures generated during execution, while the supplementary material documents the correspondence between the elements reported in the manuscript and the underlying digital files.

Consequently, the results presented in the next section are supported by specific computational evidence and can be directly traced in the reproducible repository and the associated supplementary material.

5. Results

5.1. DSI Behavior and Its Relationship with the Learning Rate

The first objective of the analysis was to verify whether the proposed criterion captures useful patterns in early training dynamics. In the main MNIST experiment, the DSI value exhibited a systematic variation as the learning rate increased, with a clear improvement in the index when leaving excessively conservative values, reaching its best values in an intermediate-to-high region of the grid, and subsequently deteriorating at the most aggressive explored extreme. This behavior is consistent with the intuition underlying the method and suggests that the early dynamic signal contains useful information for ordering the grid of candidate learning rates. As shown in Figure 1, the DSI does not evolve arbitrarily across the grid, but instead delimits a region of relatively favorable values associated with more convenient trajectories from the dynamic standpoint.

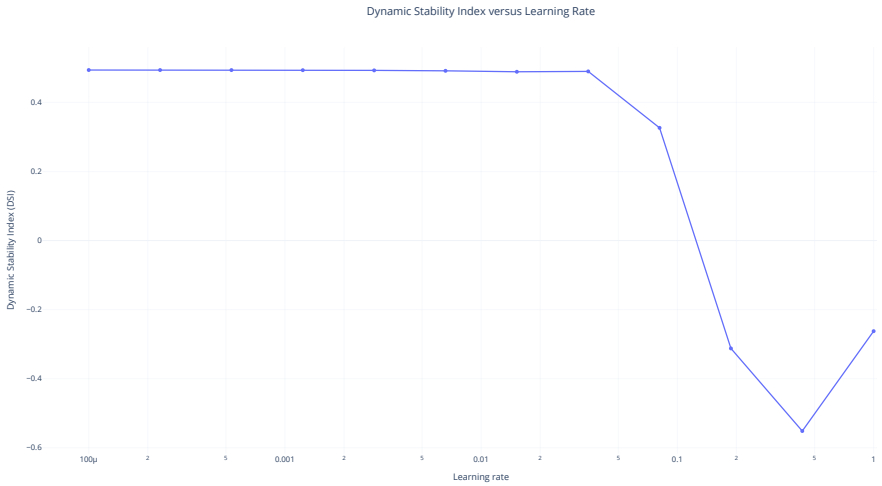

Figure 1: Evolution of the DSI as a function of the learning rate on MNIST.

The coherence of this pattern is reinforced when compared with validation performance. In particular, Figure 2 shows that the region in which the DSI improves coincides with higher validation accuracy values. This supports the usefulness of the index as an early selection criterion. On MNIST, the best performance within the main grid was obtained for $\eta = 0.432876$, with a validation accuracy of 0.8780. This result indicates that the index can identify a favorable region of the search space from a short exploratory window, without fully training all candidate configurations.

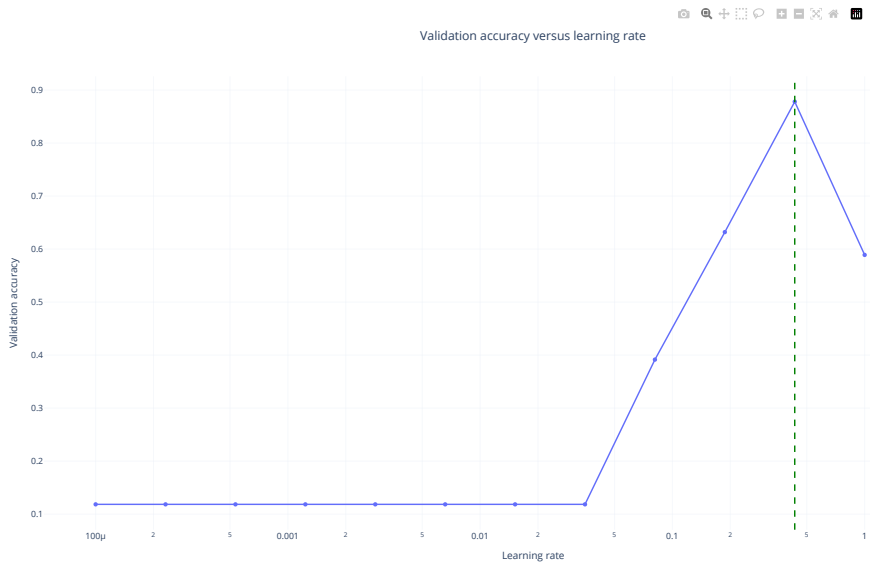

Figure 2: Validation accuracy as a function of the learning rate on MNIST.

The robustness of this selection is further reinforced by the weight-sensitivity analysis. When the criterion was recalculated over the predefined grid of combinations for α, β, γ, and δ, the main rule selected the same learning rate in all 81 evaluated configurations. This result suggests that the decision induced by the DSI does not critically depend on a single parameterization of the index, but rather reflects a relatively stable dynamic structure of the training trajectory. The complete results of this analysis are included in the supplementary material and in the associated reproducible repository.

5.2. Main Comparison Against Reference Methods

The main comparison was carried out on FashionMNIST using a lightweight CNN and contrasting the dynamic criterion against grid search, early-loss sweep, and random search. In this experiment, the central finding is that the proposed criterion matched the final performance of grid search with substantially lower exploration cost, although it did not surpass all compared methods in terms of final accuracy. As summarized in Table 1, both methods selected the same learning rate ($\eta = 0.015199$) and achieved the same final

test accuracy (0.8476); however, the dynamic criterion required less search time and a considerably smaller number of exploratory batches.

Table 1: Main comparison between methods on FashionMNIST.

Method	Learning rate	Test accuracy	Search time (s)	Search batches	Accuracy/s
Grid Search	0.015199	0.8476	26.01	1200	0.0326
Dynamic Criterion	0.015199	0.8476	10.67	360	0.0794
Early-loss Sweep	0.002848	0.8781	5.17	192	0.1697
Random Search	0.006579	0.8687	8.80	400	0.0987

In efficiency terms, the dynamic criterion required 10.67 seconds and 360 batches, compared with 26.01 seconds and 1200 batches for grid search. This result supports the central motivation of the paper: reducing learning-rate selection cost without necessarily degrading final performance.

The comparison with the other methods helps qualify the scope of the contribution. In this experiment, the early-loss sweep achieved higher final accuracy (0.8781), while random search reached 0.8687. Therefore, the dynamic criterion does not universally dominate all alternatives in terms of final predictive accuracy. Its contribution lies instead in offering a favorable combination of interpretability, low exploration cost, and closeness to exhaustive-search performance.

The relationship between DSI and validation performance reinforces this interpretation. As shown in Figure 3, lower index values tended to be associated with higher validation accuracy, suggesting that the early dynamics captured by the criterion contain useful information for discriminating more promising learning-rate configurations.

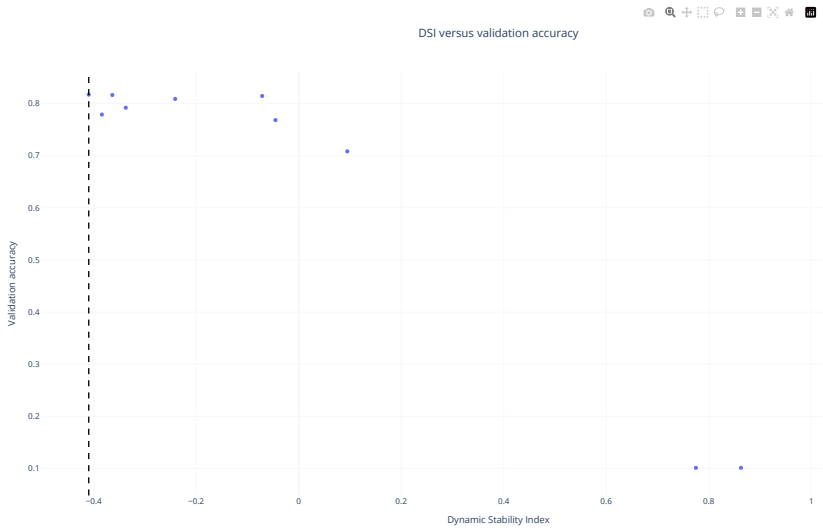

Figure 3: Relationship between DSI and validation accuracy on FashionMNIST.

This association is also reflected in the computed correlations. As reported in Table 2, the Pearson correlation between DSI and validation accuracy was -0.9679, whereas the Spearman correlation reached -0.9002. Both values indicate a strong negative relationship between the index and validation performance, reinforcing the empirical plausibility of the proposed criterion as an early mechanism for ranking the search grid.

Table 2: Correlation between DSI and validation accuracy on FashionMNIST.

Statistic	Value
Pearson correlation	-0.9679484557
Spearman correlation	-0.9001765118

The stability of the selection rule was also favorable in this problem. Variants based on minimum DSI, quantiles 0.25, 0.33, and 0.50, and rank-based combination converged to the same selected learning rate, with a holdout validation accuracy of 0.88375 in all cases. This result suggests that, at least in FashionMNIST, the final decision does not depend fragilely on a single

configuration of the rule, but rather emerges from a relatively stable region of the search space. Extended results for method comparison and rule sensitivity are included in the supplementary material and in the reproducible repository.

5.3. Generalization Across Multiple Datasets and Robustness to Variability

To evaluate to what extent the proposed criterion remains useful beyond a single problem, a multi-dataset study was conducted on two vision datasets and three tabular datasets. The aim of this block was to analyze whether the observed search-cost advantage is maintained in heterogeneous domains and, at the same time, to verify whether final performance remains in a competitive range relative to grid search. As summarized in Table 3, the dynamic criterion showed a consistent behavior in this sense: in several cases it reproduced or approximated the performance of exhaustive search with clearly lower exploration cost.

Table 3: Compact comparison between grid search and the dynamic criterion in the multi-dataset study.

Dataset	Grid acc.	Dynamic acc.	Δacc. (Dyn–Grid)	Grid time (s)	Dynamic time (s)	Speedup
MNIST (MLP)	0.9440	0.9472	+0.0032	16.30	6.83	2.39×
FashionMNIST (MLP)	0.8487	0.8487	0.0000	15.29	6.79	2.25×
Wine (tabular MLP)	0.9259	0.9259	0.0000	2.83	2.29	1.23×
Breast Cancer (tabular MLP)	0.9708	0.9591	-0.0117	2.78	1.48	1.88×
Digits (tabular MLP)	0.9759	0.9667	-0.0093	5.62	2.47	2.27×

In the base run, the dynamic criterion slightly outperformed grid search on MNIST with an MLP (0.9472 versus 0.9440), while reducing search time from 16.30 to 6.83 seconds. On FashionMNIST with an MLP, both methods achieved the same final test accuracy (0.8487), again with an important reduction in exploration time (6.79 versus 15.29 seconds). On Wine, both methods matched at 0.9259, whereas on Breast Cancer and Digits grid search retained a moderate advantage in final accuracy. Taken together, these results suggest that the dynamic criterion is not restricted to a single type of dataset, but rather preserves reasonably competitive behavior across domains with different statistical structure and complexity.

This pattern is consistent with the nature of the method. Because it relies on a short training window, the dynamic criterion aims to identify promising configurations from partial information. It is therefore expected that in some problems it replicates the grid-search decision, whereas in others

it deviates slightly from it. The important point is that, except in isolated cases, this deviation is not accompanied by drastic performance losses, while cost savings may be substantial. From a practical perspective, this reinforces the idea that the method should be understood as a low-cost early alternative rather than as an exact substitute for exhaustive validation in every scenario.

The stability of this behavior was further analyzed through a multi-seed scheme. As shown in Table 4, the dynamic criterion maintained performance very close to grid search on average, although the differences depended on the dataset considered. On FashionMNIST (MLP), both methods were practically indistinguishable in mean performance (0.8380 ± 0.0090 versus 0.8377 ± 0.0099), while the mean search time of the dynamic criterion was considerably lower (6.91 versus 15.83 seconds). On MNIST (MLP), grid search obtained a somewhat higher mean (0.9411 ± 0.0106 versus 0.9387 ± 0.0096), again at a substantially higher exploration cost (15.70 versus 7.29 seconds). A similar pattern was observed on Breast Cancer and Digits. Only on Wine did the mean time of the dynamic criterion fail to show an advantage over grid search, indicating that the cost benefit is not uniform when the search space and training cost are already very small.

Table 4: Multi-seed summary for grid search and the dynamic criterion (mean $\pm$ standard deviation).

Dataset	Grid acc.	Dynamic acc.	Δacc.	Grid mean time (s)	Dynamic mean time (s)
FashionMNIST (MLP)	0.8377 ± 0.0099	0.8380 ± 0.0090	+0.0003	15.83	6.91
MNIST (MLP)	0.9411 ± 0.0106	0.9387 ± 0.0096	-0.0024	15.70	7.29
Breast Cancer (tabular MLP)	0.9801 ± 0.0089	0.9789 ± 0.0128	-0.0012	3.12	1.79
Digits (tabular MLP)	0.9626 ± 0.0127	0.9585 ± 0.0104	-0.0041	5.04	2.85
Wine (tabular MLP)	0.9556 ± 0.0248	0.9333 ± 0.0101	-0.0223	2.50	2.58

From a statistical perspective, the differences between grid search and the dynamic criterion were not significant under the considered number of seeds. As reported in Table 5, the p-values of the paired t-test and the Wilcoxon test remained above conventional thresholds in all analyzed datasets. This result should be interpreted with caution: it does not imply strict equivalence between methods, but it does suggest that the observed differences are not strong enough, at this experimental scale, to support a conclusive superiority of one method over the other. In other words, the evidence supports a reading of competitiveness at lower cost rather than one of systematic dominance.

Table 5: Paired statistical comparison between the dynamic criterion and grid search in the multi-seed analysis.

Dataset	n pairs	Paired t-test p	Wilcoxon p	Paired Cohen's d
FashionMNIST	5	0.9600	1.0000	-0.0239
MNIST	5	0.2559	0.3750	0.5924
Breast Cancer	5	0.7040	1.0000	0.1826
Digits	5	0.1802	0.5000	0.7253
Wine	5	0.1778	0.5000	0.7303

The interpretation of effect size reinforces this moderate reading. On FashionMNIST, Cohen's d was practically null, consistent with the similarity between both methods in mean performance. On MNIST, Digits, and Wine, the values indicate an advantage of grid search in mean accuracy, although within a context where the dynamic criterion still offers appreciable reductions in search cost. Therefore, these results do not suggest that the proposed method universally replaces exhaustive search, but rather that it can act as a more efficient alternative when the goal is to approximate exhaustive-search performance using a fraction of the exploratory budget.

Overall, the multi-dataset and multi-seed evidence supports a moderate but important conclusion. The dynamic criterion does not universally replace grid search in terms of final accuracy, but it constitutes a lower-cost alternative with near-grid performance and reasonable stability across heterogeneous domains. This conclusion is especially relevant in scenarios where exploration budget is a limiting factor and fully training all candidate configurations is costly. Extended per-dataset results, additional comparison against random search, and the breakdown of wins, ties, and losses are included in the supplementary material and in the reproducible repository.

5.4. Real-World Applications: Raisin and NASA-CMAPSS

The usefulness of the proposed criterion was also evaluated in two applied scenarios with different characteristics: balanced tabular classification in Raisin and imbalanced binary risk classification in NASA-CMAPSS FD001. This block is particularly relevant because it makes it possible to analyze the behavior of the method outside the most conventional benchmarks and in contexts where exploration cost may have operational importance. As summarized in Table 6, the dynamic criterion again showed a favorable rela-

tionship between search cost and final performance, although with differences between the two problems.

Table 6: Compact comparison between the dynamic criterion and grid search in real-world applications.

Dataset	Selection metric	Grid acc.	Dynamic acc.	Grid bal. acc.	Dynamic bal. acc.	Time speedup	Batch speedup
Raisin	Validation accuracy	0.8815	0.8815	0.8822	0.8821	1.68×	3.00×
NASA-CMAPSS	Validation balanced accuracy	0.9040	0.9315	0.9072	0.8978	10.57×	22.60×

On Raisin, the dynamic criterion achieved the same final test accuracy as grid search (0.881481), with almost identical balanced accuracy (0.882133 versus 0.882243). At the same time, it did so with substantially lower search cost: 2.81 seconds versus 4.70 seconds, and 100 batches versus 300. This result provides particularly clear validation of the method, since it reproduces the performance of the exhaustive reference with a fraction of the exploratory budget.

The NASA-CMAPSS case showed a more demanding pattern from an interpretive standpoint. In this problem, the dynamic criterion selected a higher learning rate than grid search and achieved higher final test accuracy (0.931462 versus 0.904047), but somewhat lower balanced accuracy (0.897798 versus 0.907241). At the same time, the reduction in search cost was very large: from 35.70 to 3.38 seconds and from 6780 to 300 batches, corresponding approximately to a factor of 10.57 in time and 22.6 in batches.

This result should not be interpreted as an inconsistency of the method, but rather as the manifestation of a different trade-off between metrics in an imbalanced problem. Grid search, selected on balanced accuracy, favored a more conservative configuration with respect to the minority class. The dynamic criterion, in contrast, led to a solution with better global accuracy and much lower search cost, although with a slight decrease in balanced accuracy. In other words, because the dynamic criterion does not use balanced accuracy as an explicit signal during the exploratory stage, it is not expected to necessarily reproduce the same solution favored by exhaustive search optimized on that metric.

The interpretation of this difference becomes clearer when examining the classification metrics for the positive class. In NASA-CMAPSS, the configuration selected by the dynamic criterion reached a precision of 0.7383, a recall of 0.8495, and an F_1 score of 0.7900 for the risk class. These values indicate that, even without exactly maximizing balanced accuracy relative to

grid search, the selected configuration retains meaningful capacity to detect positive cases, which is important in a predictive-maintenance and early-diagnosis context.

The confusion matrices reinforce this reading. In Raisin, the dynamic solution maintained an adequate balance between both classes, with 54 true negatives, 65 true positives, 14 false positives, and 2 false negatives. In NASA-CMAPSS, the confusion matrix showed 2459 true negatives, 395 true positives, 140 false positives, and 70 false negatives, suggesting that the improvement in global accuracy did not come at the cost of collapsing minority-class detection, but rather from a redistribution of the trade-off between sensitivity and specificity.

Overall, the real-world applications reinforce the usefulness of the dynamic criterion as an early selection mechanism in applied scenarios. In Raisin, the method almost exactly reproduced the behavior of exhaustive search at lower cost. In NASA-CMAPSS, it showed that reducing exploration budget may be accompanied by a solution with better global accuracy, although not necessarily with the best balanced accuracy. This difference does not weaken the proposal; on the contrary, it more precisely delineates the type of trade-off that the criterion may induce in imbalanced and high-cost problems. The confusion matrices, complete comparison tables, extended metrics, and associated classification reports for both applications are included in the supplementary material and in the reproducible repository.

5.5. General Discussion

Taken together, the results show that the proposed dynamic criterion can exploit early training signals to select competitive learning-rate configurations at substantially lower search cost than exhaustive search. This trend is observed in the main FashionMNIST comparison, in the multi-dataset study, and, particularly clearly, in the Raisin application, where the method reproduced grid-search performance with a fraction of the exploratory budget.

At the same time, the evidence indicates that the method does not constitute a universally dominant solution. In some scenarios, simpler strategies such as early-loss sweep achieved higher final accuracy, while in others grid search preserved specific advantages, especially when the prioritized selection metric does not directly coincide with the signal used during the exploratory stage. In this sense, the dynamic criterion should be interpreted as an early, interpretable, and computationally lightweight alternative for reducing exploration cost, rather than as an exact replacement for exhaustive validation.

One of the most relevant findings of the analysis is that the DSI synthesizes, in a single measure, information about volatility, oscillation, early progress, and numerical instability. Moreover, the stability observed in the sensitivity analyses and in the robustness of the selection rule suggests that the captured signal does not merely reflect an arbitrary parametrization, but instead corresponds to empirical regularities in early training dynamics across the analyzed scenarios.

From a practical point of view, the usefulness of the criterion depends on the experimental context. In lower-cost problems, the advantage over grid search may be moderate; in contrast, in scenarios where exploring multiple complete configurations is more expensive, the reduction in exploratory budget becomes more relevant. Within this framework, the method occupies an intermediate position between exhaustive validation and simpler early criteria, combining interpretability, low exploratory cost, and competitive performance across a variety of settings.

6. Conclusions

This work addressed the problem of learning-rate selection from a perspective centered on early training dynamics. To this end, an early-selection criterion based on a Dynamic Stability Index (DSI) was proposed, built from empirical volatility, oscillation, normalized descent, and numerical instability, together with a decision rule aimed at favoring relatively stable configurations with effective early progress.

In general terms, the results indicate that a signal constructed from initial training dynamics can be used to select competitive learning rates with substantially lower exploration cost than exhaustive search across diverse scenarios. In this sense, the empirical evidence supports the central hypothesis of the study: the early stages of training contain useful information for identifying favorable configurations without fully training all alternatives.

This support, however, should be interpreted in a bounded way. The method did not show universal dominance over all references and, in some scenarios, other strategies retained specific advantages in terms of final accuracy or adequacy to specific metrics. Therefore, the proposed criterion should be understood as an early, interpretable, and computationally efficient alternative for reducing exploration cost rather than as an exact substitute for exhaustive validation.

As future work, it would be relevant to study variants of the index adapted to specific metrics, analyze its integration with broader search strategies, and examine its behavior across additional architectures, optimizers, and tasks. Overall, the presented evidence suggests that early training dynamics can be effectively exploited to construct simple, interpretable, and computationally efficient learning-rate selection rules.

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